

Money & Banking

Introduction to money

Money is an item which people use to buy and sell goods and services. It also carries out a number of other functions (see below). Bank notes and coins (cash) are used mainly for small purchases. Money in bank deposits is the main form of money. It is transferred from one person to another person by direct debit, credit cards and smartphones.

In recent years, there has been the development of what is known as cryptocurrencies (also called digital money and electronic money). These cryptocurrencies allow people to make and receive payments online, person to person, without the need for a commercial bank. They are also not regulated by central banks. The best known and most widely used cryptocurrency is bitcoin. This was introduced in 2009 by an unknown entrepreneur. People can get bitcoins by accepting them in payment, buying them on special websites or carrying out tasks for the bitcoin system (known as mining). As well as using cryptocurrencies to make payments, some people get them hoping that they will rise in value. However, their value is determined by demand and supply and their value has fluctuated considerably.

While some people use bitcoins to make payments, borrow and lend and as a way of saving, the extent to which they can be considered to be money is debatable. This is because they have not yet become generally acceptable. The majority of shops and firms do not accept them as a way of paying for products.

The functions of money

The best-known function of money is to buy and sell goods and services, but it also carries out three other functions. Each of these functions has a technical name.

- **Medium of exchange.** We sell goods and services for money. We use the money we receive to buy other goods and services. Having money overcomes the need for the double coincidence of wants. Households and firms do not have to find someone who both wants what they want to exchange and has something they want in return. For example, a farmer can sell her onions to one person and use the money she receives to buy curtains from another person.
- **Store of value.** Money enables people to save. They can keep any money they receive from selling goods and services for future use.
- **Unit of account.** This function is also known as a measure of value. Money enables the value of different items to be compared as prices are expressed in money terms. For instance, if a book sells for \$30 it can be said to be worth 15 times as much as a soft drink that sells for \$2 a can.
- **Standard of deferred payment.** Money enables people, firms and the government to borrow and lend and to buy and sell in the future. For example, someone who borrows money now can agree how much they will repay in five years' time and a trader can agree to buy a given quantity of pepper in two years' time.

There is no such thing as luck, believe in hard work!

Money & Banking

The characteristics of money

To act as money, an item has to possess a number of characteristics:

- **Generally acceptable.** This is the most important characteristic. If people are not prepared to accept an item in exchange for products, as a store of value and standard of deferred payments, if it will not act as money. In Zimbabwe, people lost confidence in the Zimbabwean dollar in 2009 due to hyperinflation (see Section 20.2). People used US dollars instead. The Zimbabwean central bank stopped printing the Zimbabwean dollar in 2009 and did not issue a new version until 2019.
- **Recognizable.** People have to identify the item as money. Central banks make the country's notes and coins distinctive.
- **Portable.** It must be easy to carry money around. Bank notes are very light and coins are not too heavy. Transferring money from one account to another account can now be done electronically and so does not involve having to carry anything. Even using credit and debit cards to transfer money involves carrying very light objects.
- **Divisible.** Any form of money must be divisible into units of different value. For example, the US currency has the following denominations: bank notes of \$1, \$2, \$5, \$10, \$20, \$50 and \$100 and coins of 1 cent (penny), 5 cent (nickel), 10 cent (dime), 25 cent (quarter) and 50 cent (half dollar) (see Figure 44.3).
- **Homogeneous.** Each unit of money must be identical. In the past, some governments called in money and reduced, for example, the silver in the coins. People held back some of their money. When the government returned the money, people valued it less than the money they kept back.
- **Limited in supply.** If there is an unlimited supply of money, it would have no value. This would make it unacceptable.
- **Not easy to counterfeit.** If it was easy to counterfeit money, it would lose its value. Central banks now build in special features to their bank notes to try to prevent counterfeiting.

The money supply

The money supply is the total amount of money in an economy. The money supply consists of currency in circulation plus relevant deposits. Governments measure the money supply to gain information about trends in aggregate demand, the state of financial markets and to help them in determining the direction of monetary policy.

In practice, measuring the money supply is not straightforward. This is because it is difficult to decide what to include in any measure of the money supply.

Economists define items as money if they fulfil the functions of money. However, the extent to which items carry out these functions varies and can change over time. As a result, governments use a variety of measures of the money supply, which are occasionally altered to reflect developments in the roles carried out by particular items. There are two main measures of the money supply:

There is no such thing as luck, believe in hard work!

Money & Banking

1. Narrow money: This is money that is used as a medium of exchange and consists of notes in circulation and cash held in banks and in balances held by commercial banks at the central bank. This is sometimes referred to as the monetary base.
2. Broad money: This consists of the items in narrow money plus a range of items that are concerned with money's functions as a store of value. An example is money in savings accounts.

The quantity theory of money

Aggregate demand is influenced by the money supply and, in turn, can influence the money supply. One theory which seeks to explain how changes in the money supply can have an impact on the economy is the quantity theory of money. This theory is based on the Fisher equation: $MV = PT$ (also commonly written as $MV = PY$) where: M is the money supply V is the velocity of circulation (the number of times money changes hands) P is the price level T or Y is the transactions or output of the economy.

Both sides of the equation have to equal each other since both sides represent total expenditure in the economy. To turn the equation into a theory, monetarists assume that V and T are constant, not being affected by changes in the money supply, so that a change in the money supply causes an equal percentage change in the price level. For example, the money supply may initially be \$80 billion, the velocity of circulation 5, price level 100 and output 4 billion. If V and T are unchanged, an increase in the money supply by 50% to 120 would cause the price level to also rise by 50% to 150. The monetarist view is that inflation is a monetary phenomenon. However, Keynesians argue that the equation cannot be turned into a theory since V and T can change with a change in the money supply and so no predictions can be made about what effect a change in M will have on P.

Keynesian and monetarist theoretical approaches

Keynesians are economists whose ideas are based on the work of the British economist John Maynard Keynes (1883-1946). They believe that if left to market forces, there is no guarantee that the economy will achieve a full employment level of GDP. Indeed, they think that the level of GDP can deviate from the full employment level by a large amount and for long periods. In such cases, they favor government intervention to influence the level of economic activity. They argue that if there is high unemployment, the government should use a budget deficit to increase aggregate demand. They believe that a government can assess the appropriate amount of extra spending to inject into the economy in such a situation. For most Keynesians, the avoidance of unemployment is a key priority.

Functions of commercial banks

Commercial banks, also called high street banks and retail banks, carry out a range of functions. They provide deposit accounts for their customers; there are two main types of deposit accounts. One type is a demand deposit account (also known as a current account or sight account). This type of account provides easy and quick access to the money in the account. It is used mainly to receive and make payments. Customers can pay in and take out cash from this type of account. The second

There is no such thing as luck, believe in hard work!

Money & Banking

type of account is known as a savings deposit account. This type of account is used mainly as a way of saving.

As well as deposits, commercial banks hold a range of other assets. These include cash, government securities and equities. Commercial banks make most of their profits from lending. Bank customers are usually given the opportunity to overdraw their deposit account by a certain value. They can also ask for an overdraft. An overdraft allows customers to spend more than is in their deposit accounts. It is often used for unexpected differences in spending and income. The customer is charged interest on the overdraft. A loan has to be agreed with a commercial bank. It is usually for a particular purpose and a set amount of time. Interest has to be paid on the full value of the loan, whether it is used or not. The interest rate, however, is normally lower than on an overdraft.

The proportion of liquid assets to total deposits that commercial banks keep is known as the reserve ratio. To learn more about the reserve ratio and capital ratio, see Section 44.5.

The objectives of a commercial bank

Commercial banks have three main objectives: profitability, liquidity and security. Commercial banks aim to achieve high profits for their shareholders mainly by lending. This objective conflicts with its two other objectives and a balance has to be achieved between the three objectives. Commercial banks have to achieve a certain level of liquidity and security. They need to have enough liquid assets, such as cash and short-term securities that will soon be turned into cash, to meet the expected request by their customers to withdraw their money in cash. These liquid assets are, however, not very profitable. Indeed, no money is earned on holding cash. Commercial banks also aim for security. They do not want to go out of business and they have to convince their customers that they are financially sound. To achieve this aim they try to ensure they have sufficient financial capital to cover their riskier loans.

A commercial bank will keep some assets that are liquid but not very profitable and some that are profitable but not very liquid. It may also keep some of its profits in case some of its borrowers do not repay their loans or some of the firms it buys shares in go out of business.

Causes of changes in money supply

There are five main causes of an increase in the money supply in an open economy:

1. an increase in commercial bank lending
2. an increase in government spending financed by borrowing from commercial banks
3. an increase in government spending financed by borrowing from the central bank
4. the sale of government bonds to private sector financial institutions (quantitative easing)
5. More money entering than leaving the country.

Commercial banks and credit creation

When commercial banks lend, they create money. This is because when a bank gives a loan (also called an advance by bankers), the borrower's account is credited with the amount borrowed. Banks

There is no such thing as luck, believe in hard work!

Money & Banking

are in a powerful position to create money because they can create more deposits than they have cash and other liquid assets (items that can be quickly converted into cash).

From experience, banks have found that only a small proportion of deposits are cashed. When people make payments, they tend to make use of credit cards, debit cards and online transfers. These means of payment involve a transfer of money using entries in the records that banks keep of their customers' deposits rather than by paying out cash.

So, banks can create more deposits than they have liquid assets. Nevertheless, they have to be careful when calculating what liquidity ratios (the proportion of liquid assets to total liabilities) to keep. This ratio can be set by a central bank when it is called a reserve ratio. The lower the proportion of liquid assets commercial banks keep, the more they can lend. However, they have to be able to meet their customers' demands for cash. If they miscalculate and keep too low a ratio, or if people suddenly start to cash more of their deposits, there is a risk of a run on the banking system. This was evidenced in 2008 when there were fears over the liquidity of some US and European banks, the result being enforced government involvement. Indeed, banking is based on confidence. Customers have to believe there is enough cash and liquid assets to pay out all their deposits even though, in practice, this is not going to be the case.

The bank credit multiplier

By estimating what reserve ratio to keep, a bank will be able to calculate its bank credit multiplier. When calculating its bank credit multiplier, a commercial bank will take into account its reserve ratio:

Bank credit multiplier (after loans have been made) = Value of new assets created / Value of change in liquid assets

For example, if total deposits rise by \$600 million as a result of a new cash deposit of \$100 million, the bank credit multiplier is $\$600\text{m} / \$100\text{m} = 6$.

It is also possible to calculate the bank credit multiplier, in advance:

Bank credit multiplier (in advance) = $100 / \text{liquidity ratio}$

Reserve ratio: If a bank keeps a reserve ratio of 5%, the bank credit multiplier will be $100 / 5 = 20$. With this knowledge, a bank can then calculate how much it can lend. The bank first works out the possible increase in its total liabilities. This is found by multiplying the change in liquid assets by the bank credit multiplier. So, if the bank credit multiplier is 20 and liquid assets rise by \$40 million, total deposits can rise by $\$40 \text{ million} \times 20 = \800 million .

To work out the change in loans (advances), the change in liquid assets is deducted from the change in liabilities. This is because the change in liabilities will include deposits given to those putting in the liquid assets. In the example, the change in loans can be $\$800 \text{ million} - \$40 \text{ million} = \$760 \text{ million}$.

However, in practice a bank may not lend as much as the bank credit multiplier implies it can. This is because there may be a lack of households and firms wanting to borrow or a lack of credit-worthy

There is no such thing as luck, believe in hard work!

Money & Banking

borrowers. If banks persist in lending to borrowers with poor credit ratings, as was the case in the US sub-prime market in 2008, the risk of default is high and can have serious consequences on a bank's liquidity.

A bank is likely to change its reserve ratio if people alter the proportion of their deposits they require in cash, if other banks alter their lending policies or if the country's central bank requires banks to keep a set reserve ratio.

A central bank may seek to influence commercial banks' ability to lend. For example, the bank may engage in open market operations. These involve the central bank buying or selling government securities to change bank lending. If the central bank wants to reduce bank loans, it will sell government securities. The purchasers will pay by drawing on their deposits in commercial banks and so cause the commercial banks' liquid assets to fall.

The capital ratio: The capital ratio is a commercial bank's available financial capital expressed as a percentage of its riskier assets. The available financial capital includes its retained profits and newly issued shares. While government securities are normally assumed to carry no risk, some loans may be considered risky if the economy got into difficulties. For example, if a commercial bank had a capital ratio of 8%, this would mean that 8% of its riskier assets are covered by readily available financial capital. The higher the capital ratio a commercial bank has, the more unexpected losses it can experience without going out of business. The capital ratio is designed to protect the bank's customers in the event of a financial crisis and to promote the stability of the banking sector by discouraging excessive risk taking.

The role of a central bank

The central bank of a country carries out a range of functions. It issues bank notes and authorizes the minting of coins. It is the bank of the commercial banks. Commercial banks keep deposits in the central bank. This enables the commercial banks to make and receive payments from other commercial banks and to withdraw money when needed. Their deposits at the central bank are considered to be liquid assets. The central bank will also usually lend to commercial banks if they get into financial difficulty.

Two of the other key functions of a central bank are to act as the banker to the government and to implement the government's monetary policy.

Government deficit financing

If the government spends more than it raises in taxation, it will have to borrow. This is organized by the central bank. If the central bank borrows, on behalf of the government, by selling government securities, including National Savings certificates, to the non-bank private sector (non-bank firms and the general public), it will be using existing money. The purchasers will be likely to draw money out of their bank deposits. So, the rise in liquid assets resulting from increased government spending will be matched by an equal fall in liquid assets as money is withdrawn.

However, if a budget deficit is financed by borrowing from commercial banks or the central bank itself, the money supply will increase. When a government borrows from the central bank, it spends

There is no such thing as luck, believe in hard work!

Money & Banking

money drawn out of its account kept at the central bank. This spending will increase deposits in commercial banks and so increases their liquid assets. With more liquid assets, they will be able to lend more. Commercial banks will also be able to lend more if the government borrows from them by selling them short-term government securities. This is because these securities count as liquid assets and so can be used as the basis for loans.

Quantitative easing: When the rate of interest is very low, a central bank may decide to try to increase aggregate demand by in the use of quantitative easing. This involves a central bank buying both government and private securities, of different maturities, from financial institutions, including commercial banks. In return for the securities, the central bank credits the accounts of the commercial banks. With more liquid assets, it is hoped that the commercial banks will lend more. This should increase the money supply and reduce the short-term and long-term interest rate. These changes may, in turn, increase investment and consumer expenditure and so aggregate demand and economic activity. Quantitative easing has also been used to help markets that are in difficulties. For example, the Federal Reserve Bank of the USA has purchased mortgage-based securities to help the US housing market.

Quantitative easing was first used in Japan in the 1990s and in the UK and USA following the global recession at the end of the first decade of the twenty-first century. It has also been used in the twenty-first century when very low, or even negative interest rates, have limited central banks' ability to stimulate economic activity by reducing the rate of interest it sets.

ZARAK MUSHTAQ

There is no such thing as luck, believe in hard work!